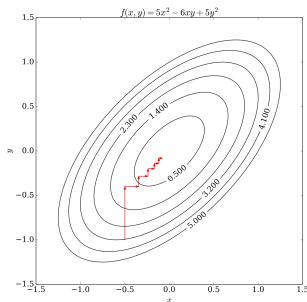
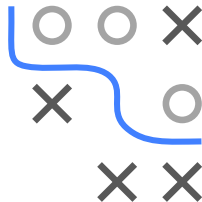


Coordinate descent

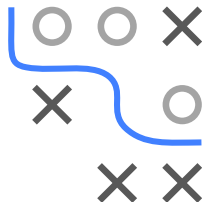


- Axes as descent direction
- Convergence criteria of CD
- CD on linear model and Lasso: soft thresholding



COORDINATE DESCENT

- Idea: instead of gradient, use coordinate directions for descent
- No gradient needed: objective function may be non-differentiable



Approach:

- Select starting point $\mathbf{x}^{[0]} = (x_1^{[0]}, \dots, x_d^{[0]})$
- Step t : Minimize f along x_i for each dimension i for fixed $x_1^{[t]}, \dots, x_{i-1}^{[t]}$ and $x_{i+1}^{[t-1]}, \dots, x_d^{[t-1]}$
- Minimum is determined with (exact/inexact) line search
- Order of dimensions: cyclic $1, 2, \dots, d$, or any other permutation

CONVERGENCE CRITERIA

- CD stops at a point that is minimal along each coordinate axis; is this a global minimizer? [▶ Click for source](#)

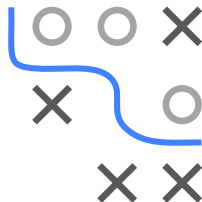
Case 1: f convex and differentiable $\Rightarrow$ **yes**

Case 2: f convex, not differentiable $\Rightarrow$ **no** in general (next slide)

Case 3: $f =$ convex differentiable + **convex separable** $\Rightarrow$ **yes**

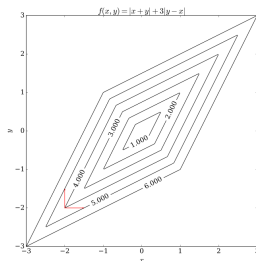
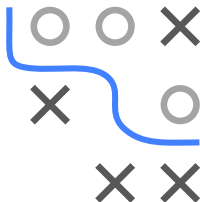
$$f(\mathbf{x}) = g(\mathbf{x}) + \sum_{i=1}^d h_i(x_i)$$

with g convex differentiable and h_i convex, possibly nonsmooth, univariate functions



EXAMPLE: NON-DIFFERENTIABLE FUNCTION

- No general convergence for convex, non-differentiable functions (case 2)
- Counterexample $f(x, y) = |x + y| + 3|y - x|$, minimum at $(0, 0)$
- At the red point, a step along either axis increases f , although both steps together would bring us closer to the optimum
- CD gets stuck at a point that is not the minimum



► [Click for source](#)

EXAMPLE: LINEAR REGRESSION

- Minimize LM with L2 loss via CD (case 1):

$$\min_{\theta} g(\theta) = \min_{\theta} \frac{1}{2} \sum_{i=1}^n (y^{(i)} - \theta^T \mathbf{x}^{(i)})^2 = \min_{\theta} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\theta\|^2$$

where $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{X} \in \mathbb{R}^{n \times d}$ with columns $\mathbf{x}_1, \dots, \mathbf{x}_d \in \mathbb{R}^n$

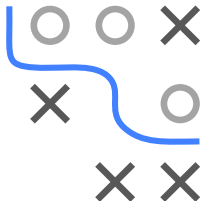
Note: smooth toy example where the minimizer along each axis is computed in closed form

- Assume orthogonal design for intuition, i.e., $\mathbf{X}^T \mathbf{X} = \mathbf{I}$:

$$g(\theta) = \frac{1}{2} \mathbf{y}^T \mathbf{y} + \frac{1}{2} \theta^T \theta - \mathbf{y}^T \mathbf{X} \theta$$

$$\stackrel{(*)}{=} \frac{1}{2} \mathbf{y}^T \mathbf{y} + \frac{1}{2} \theta^T \theta - \mathbf{y}^T \sum_{k=1}^d \mathbf{x}_k \theta_k$$

$$^{(*)} \mathbf{X}\boldsymbol{\theta} = \mathbf{x}_1\theta_1 + \mathbf{x}_2\theta_2 + \cdots + \mathbf{x}_d\theta_d = \sum_{k=1}^d \mathbf{x}_k\theta_k$$



EXAMPLE: LINEAR REGRESSION

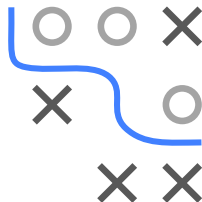
- Exact CD update in direction j :

$$\frac{\partial g(\boldsymbol{\theta})}{\partial \theta_j} = \theta_j - \mathbf{y}^T \mathbf{x}_j$$

- By solving $\frac{\partial g(\boldsymbol{\theta})}{\partial \theta_j} = 0$ we get

$$\theta_j^* = \mathbf{y}^T \mathbf{x}_j$$

- Repeat this update for all θ_j



EXAMPLE: SOFT THRESHOLDING

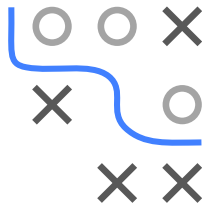
- Minimize LM with L2 loss + L1 regularization via CD (case 3):

$$\min_{\theta} h(\theta) = \min_{\theta} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\theta\|^2 + \lambda \|\theta\|_1$$

- Again we assume orthogonal design, i.e., $\mathbf{X}^T \mathbf{X} = \mathbf{I}$, and can write

$$h(\theta) = \frac{1}{2} \mathbf{y}^T \mathbf{y} + \frac{1}{2} \theta^T \theta - \sum_{k=1}^d \mathbf{y}^T \mathbf{x}_k \theta_k + \lambda \sum_{k=1}^d |\theta_k|$$

- Since $|\cdot|$ not differentiable at 0, distinguish three cases (see next slide)



EXAMPLE: SOFT THRESHOLDING

- **Case 1:** $\theta_j > 0$. Then $|\theta_j| = \theta_j$ and

$$0 = \frac{\partial h(\boldsymbol{\theta})}{\partial \theta_j} = \theta_j - \mathbf{y}^T \mathbf{x}_j + \lambda \quad \Leftrightarrow \quad \theta_{j,\text{Lasso}}^* = \theta_j^* - \lambda$$

Consistent with $\theta_j > 0$ only if $\theta_j^* > \lambda$

- **Case 2:** $\theta_j < 0$. Then $|\theta_j| = -\theta_j$ and

$$0 = \frac{\partial h(\boldsymbol{\theta})}{\partial \theta_j} = \theta_j - \mathbf{y}^T \mathbf{x}_j - \lambda \quad \Leftrightarrow \quad \theta_{j,\text{Lasso}}^* = \theta_j^* + \lambda$$

Consistent with $\theta_j < 0$ only if $\theta_j^* < -\lambda$

- **Case 3:** $\theta_j = 0$, remaining case where $|\theta_j^*| \leq \lambda$

